# A3-PS001-TC171

## Android Ransomware-like Activity Early Warning in a Controlled Lab

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# Team Introductions

## Susnata Maiti
### Team Lead

**Role**
Project coordination and detection engine

**Responsible for**
Alert rule, response, testing and results

**Contribution**
Connect the engine to the hook, evaluate events and explain the outcomes

## Moupiya Manna
### Team Member (Hook)

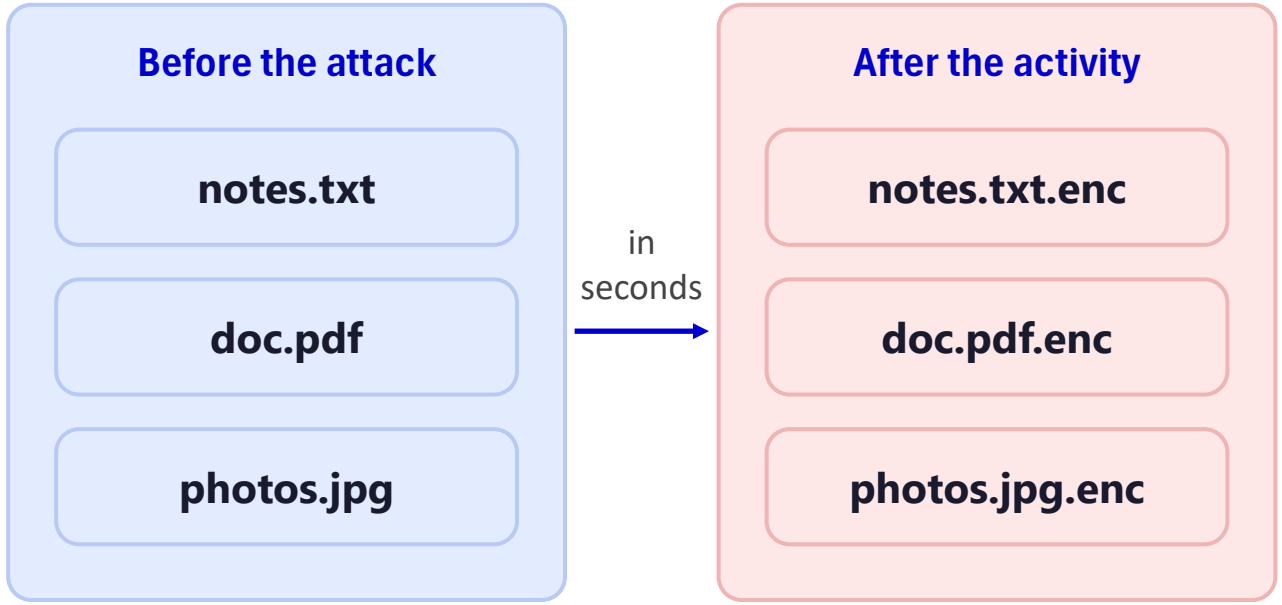
**Role**
Frida hook and event monitoring

**Responsible for**
Analysing APK file operations and observing renames

**Contribution**
Capture file-rename events and send them to the detection engine

Prof. Baisakhi Das presents the simulator workloads in the technical section.

# Problem Description




**Problem**
Zero-day ransomware has no signature, but it renames many files within seconds.

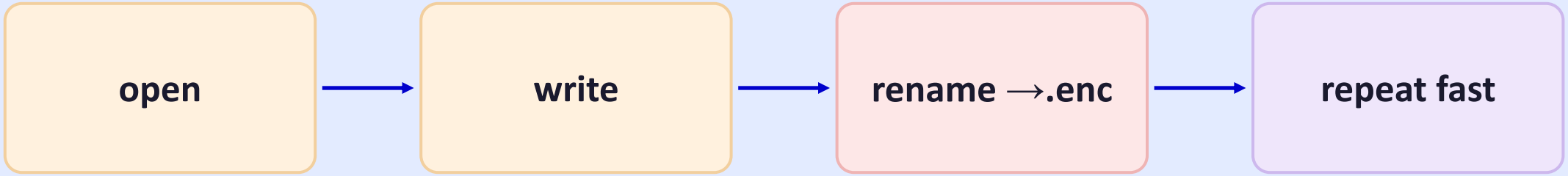
**Approach**
Hook file renames in a test directory; a rate rule flags bursts.

**Outcome**
3/3 simulated attacks detected, 0/2 false alarms (lab setting).

Goal: raise an early alert and stop only the test workload

# Background

## Ransomware file-system pattern




**Android sandbox**
An app cannot read another app's memory.

⇒

**Our observation point**
File renames inside one test directory.

Safe by design: emulator only • no real malware • no device-wide monitoring

# Literature & RFC Review

| Work | Approach | Key signals |
|---|---|---|
| **Signature antivirus** | Match known binaries | Hashes and patterns |
| **HelDroid [3]** | Static analysis of mobile ransomware | Lock-screen, encryption and threat-text features |
| **CryptoLock [4]** | Dynamic file-activity monitoring | Entropy, file-type change, similarity, deletion |
| **ShieldFS [5]** | Filesystem monitoring and recovery | I/O activity models, shadow copies |
| **UNVEIL [6]** | Automated dynamic analysis | File I/O traces, screen locking |
| **This work** | **Android hook on File.renameTo** | **Rename rate (≥ 10 in 2 s)** |

Our focus: a lightweight, explainable rename-rate signal for Android. Entropy and other signals are future work.

No IETF RFC covers app-level file-activity monitoring, so Android security guidance [1] is used.

# Proposed Solution

**`Red_Team_Simulator.apk`**
5 harmless workloads

→

**`ueba_hook.js`**
Frida hook on File.renameTo

→

**`ueba_engine.py`**
≥ 10 renames in 2 s ⇒ alert

→

**`adb force-stop`**
Test app only

## Workloads in the APK (test directory)

**CryptoLocker-style**
simulated attack

**DeepTree**
simulated attack

**PolyWiper**
simulated attack

**DocAutoSave**
benign

**PhotoSync**
benign

Ethics and safety: no live malware • isolated emulator • only the test directory is touched

# Optimization Proposed

## Design decisions

| Decision | Baseline | Ours |
|---|---|---|
| **Trigger** | Any rename | Renames in the test path |
| **Decision** | Single event | ≥ 10 events in 2 s |
| **Response** | Manual stop | Auto force-stop of test app |
| **Telemetry** | None | Delay and files changed |

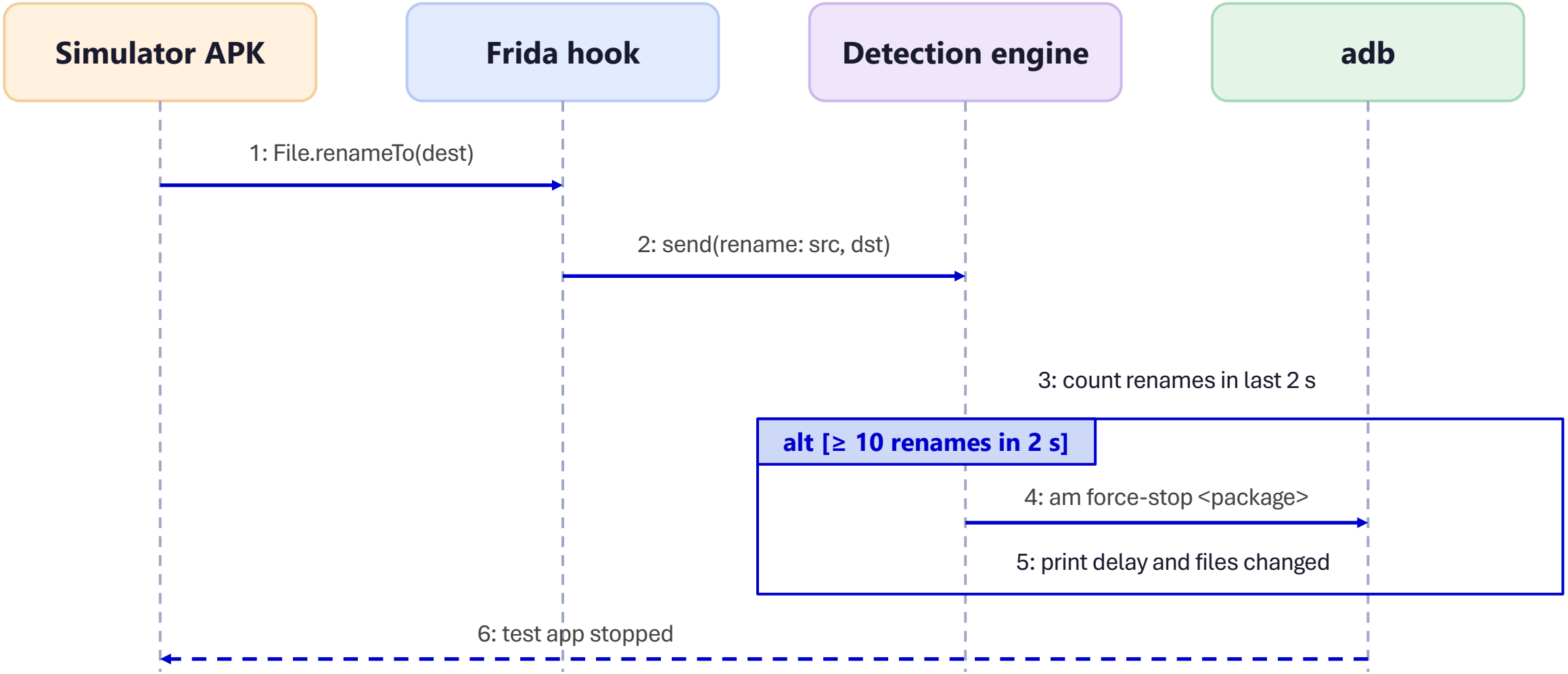
## Lab results (5 runs)

| Workload | Result | Delay | Files |
|---|---|---|---|
| **CryptoLocker-style** | Stopped | [ s ] | [ n ] |
| **DeepTree** | Stopped | [ s ] | [ n ] |
| **PolyWiper** | Stopped | [ s ] | [ n ] |
| **DocAutoSave** | Ignored | — | — |
| **PhotoSync** | Ignored | — | — |

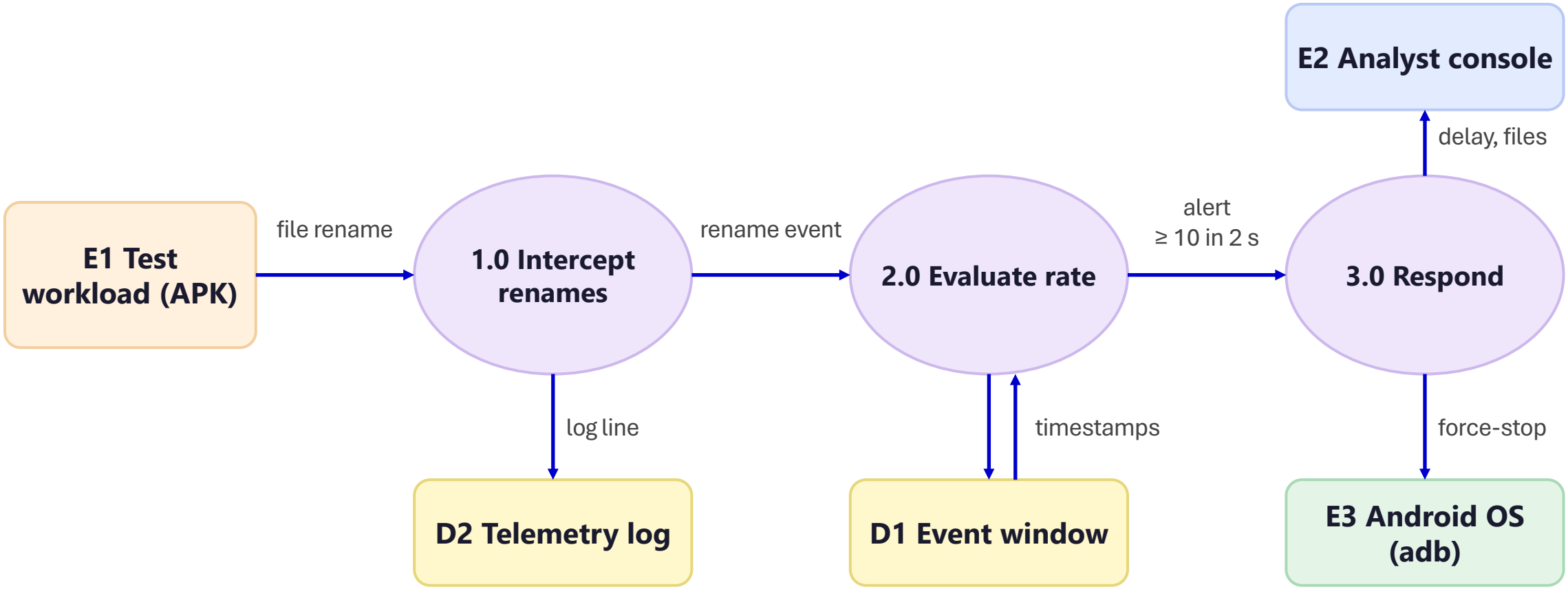
**TP 3 • TN 2 • FP 0 • FN 0**

Why 10 renames in 2 s? Benign workloads rename at most one file per 2 s, while attacks rename in bursts. 10 sits well above normal use. A threshold sweep is planned.

# Solution Architecture & Design



# Data Flow Diagram




Rectangle = external entity or data store • Circle = process • Arrow = data flow

# Deployment Timeline

**Week 1**
**Lab setup**
Emulator, Frida

→

**Week 2**
**Workloads**
3 attacks, 2 benign

→

**Week 3**
**Detector**
Hook and engine

→

**Week 4**
**Evaluate**
Delay, files, repeats

→

**Week 5**
**Demo & docs**
Repo, PDF, demo

## Known limitations (threat model)

**Low and slow**
Under 10 renames per 2 s is never flagged.

**No rename**
Overwrite, or delete-and-write, bypasses the rule.

**Native code**
libc rename() skips the Java File.renameTo hook.

**Lab setup only**
Frida and a host PC. Real use needs on-device monitoring.

Future work: libc-level hooks • file-entropy check • multi-signal score • on-device monitor

# Bibliography

[1] Android Developers, "Security best practices," developer.android.com/privacy-and-security/security-best-practices (accessed Oct. 2026).

[2] Frida, "Frida: Dynamic instrumentation toolkit," frida.re/docs (accessed Oct. 2026).

[3] N. Andronio, S. Zanero, and F. Maggi, "HelDroid: Dissecting and detecting mobile ransomware," in Proc. RAID, 2015.

[4] N. Scaife, H. Carter, P. Traynor, and K. R. B. Butler, "CryptoLock (and drop it): Stopping ransomware attacks on user data," in Proc. IEEE ICDCS, 2016.

[5] A. Continella et al., "ShieldFS: A self-healing, ransomware-aware filesystem," in Proc. ACSAC, 2016.

[6] A. Kharraz et al., "UNVEIL: A large-scale, automated approach to detecting ransomware," in Proc. USENIX Security, 2016.




# Thank You for Your Attention!

Susnata Maiti • Moupiya Manna • Prof. Baisakhi Das